

HAAPAVEST MUSTIKKAMAKI, Finland

WMO: 027970

Lat: 64.142N

Lon: 25.423E

Elev: 367

StdP: 14.50

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.7	-10.7	-20.6	1.8	-15.6	-15.1	2.5	-10.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	80.8	65.4	76.7	63.0	73.2	61.2	67.3	76.3	65.2	73.6	63.1	70.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	91.4	71.1	61.8	83.7	68.5	59.8	77.8	66.7	31.9	76.8	30.3	73.5	28.8	71.1	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-17.8	84.2	8.4	5.2	-23.9	87.9	-28.8	91.0	-33.6	93.9	-39.7	97.7		
(5)			-18.1	69.9	8.3	3.7	-24.1	72.6	-29.0	74.8	-33.7	76.8	-39.7	79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.9	14.2	18.0	24.7	35.6	48.2	55.3	61.5	57.5	49.0	36.9	29.8	22.3		
	DBStd	17.90	13.00	12.80	8.96	6.57	8.38	7.09	6.28	5.85	6.15	6.66	8.07	11.31		
	HDD50	5353	1110	897	784	432	134	26	2	8	90	406	606	858		
	HDD65	9958	1575	1317	1249	881	522	301	145	240	479	870	1056	1323		
	CDD50	927	0	0	0	1	78	186	358	242	61	1	0	0		
	CDD65	58	0	0	0	0	2	11	36	9	0	0	0	0		
	CDH74	655	0	0	0	0	60	128	376	91	0	0	0	0		
	CDH80	126	0	0	0	0	3	22	89	12	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	21.9	1.5	1.2	1.0	1.0	1.8	2.3	3.0	2.6	2.1	1.9	1.8	1.7		
(15) Precipitation	PrecMax	27.5	2.8	2.2	1.9	2.0	3.1	4.7	5.7	5.7	4.5	3.5	3.4	3.6		
	PrecMin	16.1	0.5	0.2	0.2	0.4	0.4	0.7	0.7	0.6	0.7	0.2	0.2	0.6		
	PrecStd	3.0	0.6	0.5	0.5	0.5	0.7	1.1	1.1	1.3	1.0	0.9	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	40.6	45.9	64.7	79.2	83.4	86.4	82.0	69.3	54.7	48.1	39.3		
		MCWB	36.4	37.4	39.1	49.5	59.1	64.2	67.1	68.7	60.2	51.7	46.4	38.4		
	2%	DB	34.2	37.8	41.9	55.9	75.2	77.8	82.6	76.1	64.6	51.1	44.0	36.9		
		MCWB	33.3	35.4	36.7	45.1	57.5	63.6	66.6	64.2	57.3	48.8	42.9	35.9		
	5%	DB	32.7	34.9	38.4	51.1	69.9	72.9	78.0	72.1	61.2	48.8	41.6	35.1		
		MCWB	32.0	33.8	34.7	42.9	56.0	59.5	64.5	61.9	55.5	47.0	40.9	34.4		
	10%	DB	31.0	33.0	36.1	46.9	63.9	69.0	74.3	68.4	58.7	46.4	39.3	33.6		
		MCWB	30.4	32.2	33.4	40.2	52.1	58.2	63.0	60.3	54.6	44.7	38.7	33.2		
	0.4%	WB	36.3	37.9	40.2	50.9	62.1	67.3	71.4	69.6	61.3	53.0	46.6	38.9		
		MCDB	36.7	40.0	45.1	59.1	73.7	77.1	80.8	78.5	67.3	54.3	48.1	39.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.5	35.8	37.4	46.8	59.4	65.0	68.2	66.0	58.6	49.5	42.9	36.1		
		MCDB	34.0	37.3	40.2	54.8	71.4	74.4	77.4	74.1	62.8	50.4	43.7	36.5		
	5%	WB	32.2	33.9	35.5	43.3	56.4	62.1	66.2	63.4	56.8	47.2	40.9	34.4		
		MCDB	32.7	34.8	38.0	49.4	68.6	70.6	75.1	69.4	60.1	48.5	41.6	34.8		
	10%	WB	30.4	32.3	33.8	40.9	53.2	59.3	64.2	61.6	54.9	44.8	38.8	33.2		
		MCDB	30.9	32.9	35.4	46.1	62.9	67.0	72.4	67.0	57.9	46.2	39.2	33.6		
	5% DB	MDBR	10.9	11.2	14.9	16.1	19.5	17.8	16.9	13.8	9.5	6.7	8.1			
		MCDBR	7.6	8.7	14.9	24.4	29.0	26.0	25.2	23.5	18.9	12.2	7.8	7.1		
		MCWBR	7.6	7.4	11.2	15.0	14.8	13.2	12.0	12.9	12.4	10.3	7.5	6.6		
		MCWBR	7.7	8.2	12.4	22.1	26.5	21.1	21.5	19.5	17.0	11.3	7.9	6.5		
(35) Mean Daily Temperature Range	5% WB	MCWBR	7.7	7.2	9.9	14.5	14.4	11.7	11.4	11.7	11.8	10.1	7.7	6.4		
	taub		0.214	0.279	0.308	0.348	0.344	0.347	0.368	0.350	0.327	0.306	0.355	0.356		
		taud	1.976	2.127	2.187	2.304	2.447	2.462	2.421	2.461	2.515	2.379	2.622	2.643		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		110	197	245	262	275	277	266	260	244	193	102	55		
		Edh at Noon	9	20	28	32	30	31	31	27	21	16	8	5		
	RadAvg		45	218	685	1187	1555	1721	1596	1203	680	253	59	16		
(44) All-Sky Solar Radiation	RadStd		4	41	68	108	163	147	155	120	71	44	11	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (41 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page